

MSME Synthetic Data Dictionary

This dataset is synthetically generated to simulate real-world MSME financial behavior for credit risk modeling and analytics.

1. Business Profile Features

Business_Age

- **Type:** Numeric (float)
- **Distribution:** Exponential (clipped 1–30)
- **Range:** 1–30
- **Description:** Age of the business in years. Most firms are young, few are mature.

Industry_Type

- **Type:** Categorical
- **Values:** Retail, Manufacturing, Services, Wholesale, Hospitality, Agriculture
- **Description:** Sector of MSME business

Entity_Type

- **Type:** Categorical
 - **Values:** Proprietorship, Partnership, LLP, Private Limited
 - **Description:** Legal structure of the business
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2. Financial Performance Features

Monthly_GST_Sales

- **Type:** Numeric (float)
- **Distribution:** Log-normal (clipped $2e5$ to $2e7$)
- **Description:** Monthly GST turnover proxy. Highly skewed with long tail distribution.

GST_Growth_Rate

- **Type:** Numeric (float)
- **Distribution:** Normal (age-dependent mean, clipped -0.40 to 0.60)
- **Description:** Growth rate of business. Younger firms grow faster with higher variability.

3. Compliance & Tax Behavior

GST_Filing_Delay

- **Type:** Numeric (float)
- **Distribution:** Gamma (clipped 0–180 days)
- **Description:** Delay in GST filing. Higher values indicate weaker compliance.

GST_Compliance_Rate

- **Type:** Numeric (float)
 - **Derived From:** GST filing delay + noise
 - **Range:** 0 to 1
 - **Description:** Normalized compliance score. Higher means better compliance.
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4. Transactional Behavior (UPI)

Monthly_UPI_Count

- **Type:** Integer
- **Distribution:** Poisson (industry-specific λ)
- **Description:** Number of UPI transactions per month

Monthly_UPI_Value

- **Type:** Numeric (float)
- **Construction:** $\text{UPI_Count} \times \text{stochastic ticket size (capped by sales)}$
- **Description:** Total UPI transaction value per month

Digital_Sales_Ratio

- **Type:** Numeric (float)
 - **Distribution:** Beta
 - **Range:** 0 to 1
 - **Description:** Share of digital transactions in total sales
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5. BankingRange: 0 to 1 Behavior (AA Features)

AA: Account Aggregator

Monthly_Credit

- **Type:** Numeric (float)
- **Construction:** $\text{GST_Sales} \times \text{Uniform}(0.85-1.05)$
- **Description:** Total monthly inflows into bank account including minor non-sales inflows

Monthly_Debit

- **Type:** Numeric (float)
- **Construction:** $\text{Monthly_Credit} \times \text{Expense_Ratio}$
- **Special Case:** ~10% businesses are loss-making (expense ratio > 1)
- **Description:** Total monthly outflows

Average_Bank_Balance

- **Type:** Numeric (float)
 - **Construction:** $\text{Monthly_Credit} \times \text{factor} \times \text{age adjustment}$
 - **Description:** Average monthly bank balance. Skewed distribution reflecting business size and maturity.
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6. Operational Risk Features

Cashflow_Volatility

- **Type:** Numeric (float)
- **Distribution:** Beta + early-stage adjustment
- **Range:** 0 to 1
- **Description:** Stability of cash inflows and outflows

EMI_Bounce_Count

- **Type:** Integer
 - **Distribution:** Poisson (based on cashflow volatility)
 - **Range:** 0 to 5
 - **Description:** Number of EMI payment failures indicating credit stress
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7. Workforce & Operations

Employee_Count

- **Type:** Integer
- **Distribution:** Poisson + sqrt(sales scaling)
- **Description:** Total number of employees in the business

Payroll_Consistency

- **Type:** Numeric (float)
- **Distribution:** Tanh transformation + noise
- **Range:** 0 to 1
- **Description:** Stability and regularity of salary payments

Vendor_Payment_Delay

- **Type:** Numeric (float)
- **Distribution:** Gamma + working capital effect
- **Range:** 0 to 120 days
- **Description:** Delay in payments to suppliers

Working_Capital_Cycle

- **Type:** Numeric (float)
- **Distribution:** Normal (clipped 20–180)
- **Description:** Time required to convert inventory into cash